

KAGE Fitness Studio ERP - Logical Data Model Overview

This artefact explains how memberships, PT sessions, payments, freelancer billing, equipment maintenance, and reporting connect in a simple, business-friendly way.

Project Type Independent Business Analyst Portfolio Case Study	Domain Fitness Studio / ERP / Gym Operations
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Portfolio Boundary Note

This is a fictional BA portfolio artefact inspired by real-world gym operations. It is not a live client project, production ERP system, or final technical database design. The purpose is to demonstrate BA thinking and data modelling in a recruiter-friendly format.

1. DOCUMENT PURPOSE

Why This Artefact Exists

This document explains how the main business data in KAGE Fitness Studio ERP connects across daily gym operations. It supports BA portfolio storytelling by showing how business workflows can be translated into structured data relationships without becoming overly technical.

2. DATA MODEL SUMMARY

What the Model Stores

Business view

The model stores member profiles, membership plans, payments, PT packages, PT sessions, trainers, freelancer billing, equipment maintenance, and revenue reporting. It gives the studio one connected structure instead of separate spreadsheets, notes, and manual follow-ups.

Why it matters

Each entity exists because it supports a real business need such as renewals, PT balance tracking, freelancer access billing, or repair history. The result is better operational control, billing accuracy, and management visibility.

3. ENTITY OVERVIEW TABLE

Core Entities

Entity Name	Business Purpose	Key Data Stored	Connected To	Why It Matters
Member	Stores all gym users in one central profile.	Member ID, name, phone, email, member type, plan, join date, status.	MembershipPlan, Payment, PTPackage, PTSession, FreelancerClient.	Supports access, billing, renewals, PT activity, and reporting.
MembershipPlan	Stores available membership options.	Plan name, duration, base fee, negotiable status, active/inactive status.	Member.	Ensures the correct fee and expiry logic are applied.
Payment	Stores payment and invoice details.	Amount, payment purpose, mode, status, invoice number, shared status, payment date, collected by.	Member, RevenueReport.	Supports receipts, dues tracking, and revenue visibility.
Trainer	Stores trainer details and availability-related data.	Name, specialisation, shift status, commission rate, status.	PTSession.	Helps with PT allocation and workload visibility.
PTPackage	Stores personal training package purchases.	Package type, total sessions, completed sessions, remaining sessions, status.	Member, PTSession, Payment.	Tracks PT balance and progress clearly.
PTSession	Stores each individual PT session.	Date, time, session status, trainer, member, package reference, commission status.	Member, Trainer, PTPackage.	Tracks attendance, completion, cancellations, and eligibility.

3. ENTITY OVERVIEW TABLE

Core Entities - Continued

Entity Name	Business Purpose	Key Data Stored	Connected To	Why It Matters
Freelancer	Stores freelancer access and billing details.	Freelancer ID, linked member ID, monthly access fee, active client count, payment status, gate flag.	Member, FreelancerClient, FreelancerBilling.	Separates facility billing from private coaching income.
FreelancerClient	Stores clients linked to a freelancer.	Client ID, linked member ID, freelancer ID, package type, active status.	Member, Freelancer.	Helps calculate monthly freelancer billing.
FreelancerBilling	Stores monthly freelancer billing calculations.	Billing month, own fee, active client count, per-client fee, monthly payable, payment status.	Freelancer, RevenueReport.	Supports transparent monthly billing and gate-flag decisions.
Equipment	Stores gym equipment records.	Equipment ID, name, category, location, current status, purchase date.	MaintenanceLog.	Supports asset tracking and maintenance ownership.
MaintenanceLog	Stores repair and maintenance events.	Repair date, issue description, repair cost, vendor, resolution status, next service date.	Equipment, RevenueReport.	Helps review equipment condition and repeated repairs.
RevenueReport	Summarises operational performance.	Membership revenue, PT revenue, freelancer revenue, maintenance cost, net revenue, report period.	Payment, FreelancerBilling, MaintenanceLog.	Gives management a consolidated business view.

4. RELATIONSHIP EXPLANATION

How the Data Connects

MembershipPlan -> Member

One plan can be linked to many members, helping the ERP apply the correct fee and expiry logic.

Member -> PTSession

Each PT session links back to the member so attendance, completion, and cancellations can be tracked.

Freelancer -> FreelancerBilling

Monthly billing records capture the freelancer access-fee logic across billing periods.

Member -> Payment

One member can have many payments for membership, renewals, PT packages, or other charges.

Trainer -> PTSession

A trainer can conduct many PT sessions, supporting workload visibility and allocation control.

Equipment -> MaintenanceLog

One equipment item can have many maintenance records, giving a clear repair history.

Member -> PTPackage

A member can purchase more than one PT package over time, supporting package history.

Freelancer -> FreelancerClient

One freelancer can have many clients, helping calculate monthly billing.

RevenueReport summarises data

RevenueReport is a reporting view that combines operational data from payments, freelancer billing, and maintenance logs.

5. KEY BA MODELLING DECISIONS

Why the Model Was Designed This Way

One Member entity

Supports Direct Members, Freelancers, and Freelancer Clients in one structure.

MemberType

Separates how people are treated in billing and reporting.

FreelancerBilling

Monthly and based on access-fee logic, not session-based billing.

RevenueReport

A reporting view, not a transaction table.

PTSession traceability

Linked to both member and trainer for session, workload, and commission tracking.

MaintenanceLog

Supports replacement review decisions through repair history.

6. BUSINESS RULES

Rules Supported by the Model

Rule Area	Business Rule	Data Entities Supporting It
Duplicate Member Control	A duplicate member should not be created using the same phone or email.	Member
PT Balance Control	PT package balance reduces only when a PT session is marked completed.	PTSession, PTPackage
Trainer Allocation	Admin should allocate PT clients based on availability, workload, timing, goal, and specialisation.	Trainer, PTSession, PTPackage
Freelancer Billing	Monthly payable is calculated using own fee plus active client count multiplied by per-client access fee.	Freelancer, FreelancerClient, FreelancerBilling
Freelancer Gate Flag	If payment is overdue beyond the threshold, a gate flag can be activated for review.	Freelancer, FreelancerBilling
Renewal Tracking	Renewal due and overdue member status should be visible for front-desk follow-up.	Member, MembershipPlan, Payment
Equipment Review	Repeated repairs within a defined period should trigger replacement review.	Equipment, MaintenanceLog
Revenue Reporting	Reports should show membership revenue, PT revenue, freelancer revenue, maintenance cost, and net revenue.	Payment, FreelancerBilling, MaintenanceLog, RevenueReport

7. REPORTING LOGIC

How Reports Are Derived

Membership revenue

Calculated from Payment records where the purpose relates to membership registration or renewal.

Freelancer revenue

Generated from FreelancerBilling, which keeps facility revenue separate from private coaching income.

Net revenue

Shown by combining revenue categories and subtracting expenses such as maintenance, giving management a single performance view.

PT revenue

Derived from Payment and PTPackage records, with PTSession supporting package usage tracking.

Maintenance cost

Calculated from MaintenanceLog records, including repair cost and vendor details.

8. DATA MODEL ASSUMPTIONS

What This Model Is Not

1	This is a logical model, not a physical database design.
2	Field names and relationships are simplified for BA explanation.
3	API integrations such as payment gateway, WhatsApp, and biometric attendance are outside MVP scope.
4	RevenueReport may be represented as a dashboard query, reporting view, or analytics layer.

5 | Freelancer billing uses an access-fee model. Session-based billing does not apply to freelancers in this design.